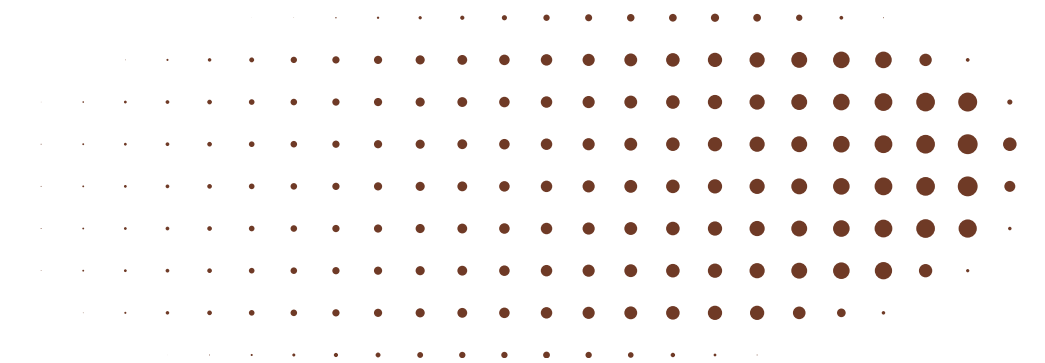


RETAIL SALES ANALYSIS :

REVENUE, GROWTH, CUSTOMER & DISCOUNT INSIGHTS (SQL)



Project Overview



01

Messy Retail Fashion Dataset

Contained Tables :

- sales_data
- product_data
- customer_data
- store_data

02

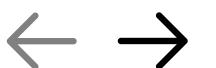
Objective :

- Analyze revenue drivers
- Understand customer behavior
- Identify growth trends

03

Tools Used :

- PostgreSQL
- Visual Studio Code
- Canva



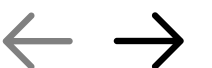
Revenue Contribution by Product Category

Valuable Insights:

- 01** Revenue is evenly distributed across major product categories, with each contributing around 19–20% of total revenue
- 02** A small portion of revenue comes from “Unknown” category, highlighting a potential data classification issue worth further investigation

	category character varying (11) 🔒	net_rev numeric 🔒	percentage numeric 🔒
1	Accessories	2236494.1570	19.91
2	Shoes	2236747.1210	19.91
3	Dresses	2218148.2210	19.74
4	Tops	2209499.1560	19.67
5	Bottoms	2184002.5760	19.44
6	Unknown	149840.4250	1.33

```
1  with category_rev as (  
2  select  
3      pd.category,  
4      sum(pd.list_price * sd.quantity * (1-sd.discount)) as net_rev  
5  from sales_data sd  
6  join product_data pd on sd.product_id = pd.product_id  
7  where returned != 1  
8  group by pd.category  
9  )  
10 select  
11     category,  
12     net_rev,  
13     round((net_rev * 100.0) / sum(net_rev) over(),2) as percentage  
14 from category_rev  
15 order by percentage desc
```



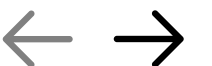
Customer Purchase Behavior by Discount Level

Valuable Insights:

- 01 The majority of transactions (63%) occur without any discount, indicating that discounts are not the primary driver of purchase decisions
- 02 This suggests relatively strong baseline pricing and product appeal without heavy reliance on discounts

	discount numeric (3,2) 🔒	total_transaction bigint 🔒	percentage numeric 🔒
1	0.30	2536	5.00
2	0.00	29907	63.00
3	0.20	4980	11.00
4	0.10	9994	21.00

```
1 select
2     sd.discount,
3     count(*) as total_transaction,
4     round((count(*) / sum(count(*) over()),2) * 100 as percentage
5 from sales_data sd
6 join product_data pd on sd.product_id = pd.product_id
7 where sd.is_discount_valid = 1
8 group by sd.discount
```



Customer Segmentation by Total Spending

Valuable Insights:

- 01 The customer base is dominated by Medium Value customers, while High Value customers represent a smaller but strategically important segment
- 02 The large Medium Value segment presents a strong opportunity for upselling and conversion into High Value customers

	customer_id [PK] character varying (7)	rev numeric	segementation text	total_customers bigint
1	C022042	1340.0800	High Value	3076
3076	C003921	1000.1250	High Value	3076
3077	C007919	999.9490	Medium Value	16400
19476	C022604	100.0800	Medium Value	16400
19477	C009067	99.9920	Low Value	1800
21277	C023931	39.2840	Low Value	1800
21278	C021511	[null]	No Value	3725

```
1 with rev_tab as
2 (
3     select
4         sd.customer_id as cust_id,
5         sum(pd.list_price * sd.quantity * (1-sd.discount)) as rev
6     from sales_data sd
7     join product_data pd on sd.product_id = pd.product_id
8     group by cust_id
9 ), segment as (
10 select
11     cd.customer_id,
12     rev,
13     case
14         when rev > 1000 then 'High Value'
15         when rev > 100 then 'Medium Value'
16         when rev <= 100 then 'Low Value'
17         else 'No Value'
18     end as segementation
19 from customer_data cd
20 left join rev_tab r on cd.customer_id = r.cust_id
21 )
22 select
23     *,
24     count(*) over(PARTITION BY segementation) as total_customers
25 from segment
```



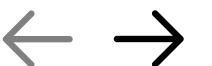
Repeat Customer Analysis

Valuable Insights :

- 01 A significant majority of customers (66.8%) have made repeat purchases, indicating a relatively strong level of customer retention
- 02 Despite the strong repeat rate, roughly one-third of customers are one-time buyers, representing an opportunity to improve retention strategies

	repeat_customers bigint	total_all_customers bigint	percentage_repeat numeric
1	14209	21276	66.78

```
1 with customer_counts as
2 (
3     select
4         customer_id,
5         count(customer_id) as total_transaction
6     from sales_data
7     group by customer_id
8 ),summary_stats as
9 (
10    select
11        count(customer_id) AS total_all_customers,
12        count(case when total_transaction > 1 then 1 end) as repeat_customers
13    from customer_counts
14 )
15 select
16     repeat_customers,
17     total_all_customers,
18     round((repeat_customers * 100.0 / total_all_customers), 2) as percentage_repeat
19 from summary_stats
```



Monthly Revenue Trend & Growth

Valuable Insights:

- 01 Monthly revenue shows noticeable fluctuations, indicating a non-linear growth pattern rather than a consistent upward trend.
- 02 Several months experience positive growth spikes (e.g., March, May, July, October), indicating periods of stronger sales momentum

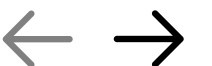
```
1 select
2     month,
3     rev as revenue_now,
4     lag(rev) over(order by month) as revenue_past,
5     round((rev - lag(rev) over(order by month)) / lag(rev) over(order by month),2) * 100 as growth_percentage
6 from
7 (
8     select
9         extract(MONTH from to_date(date,'YYYY-MM-DD')) as month,
10        sum(pd.list_price * sd.quantity * (1-sd.discount)) as rev
11    from sales_data sd
12    join product_data pd on sd.product_id = pd.product_id
13    group by month
14    order by month
15 ) as growth
```

	month	revenue_now	revenue_past	growth_percentage
	numeric	numeric	numeric	numeric
1	1	1020131.8370	[null]	[null]
2	2	983552.6030	1020131.8370	-4.00
3	3	1061775.6740	983552.6030	8.00
4	4	1032002.6230	1061775.6740	-3.00
5	5	1102926.7810	1032002.6230	7.00
6	6	1022174.3630	1102926.7810	-7.00
7	7	1087646.2150	1022174.3630	6.00
8	8	1026669.9420	1087646.2150	-6.00
9	9	1018597.8180	1026669.9420	-1.00
10	10	1060681.5890	1018597.8180	4.00
11	11	1030346.5430	1060681.5890	-3.00
12	12	1013250.7590	1030346.5430	-2.00

Key Takeaways

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Junior Data Analyst | SQL Portfolio

- 01** Revenue is evenly distributed across major product categories, indicating a healthy, diversified product portfolio that mitigates dependency risk
- 02** The majority of purchases (63%) happen without discounts, suggesting strong inherent product appeal and effective baseline pricing
- 03** A solid 66.8% repeat customer rate reflects good retention, while the dominant Medium Value segment presents a clear, high-potential audience for upselling
- 04** Revenue growth is seasonal/fluctuating, with clear high performance months (e.g., Mar, May, Jul, Oct) that warrant deeper analysis for replication



Actionable

Strategic Recommendations

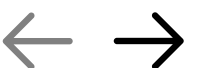
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01 Launch an "Upgrade Program" targeted at Medium Value Customers with personalized product bundles or loyalty perks

↳ Expected Impact : Convert medium spenders into high-value customers, directly boosting Customer Lifetime Value

02 Analyze the drivers behind high growth months (Mar, May, Jul, Oct) to identify and codify successful strategies for future campaigns.

↳ Expected Impact : Create a playbook to replicate success, smoothing out revenue fluctuations and planning more effective promotions.



What This Analysis Demonstrates:

- 01 Ability to translate raw retail data into clear business insights
- 02 Strong foundation in SQL analytics
- 03 Structured storytelling from data to action

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Thank You
Lets Work
Together

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